# Engineering Mathematics: Linear Algebra

## Topic: Matrix Multiplication (The Dot Product Method)

### 1. The Rule of Compatibility

Before multiplying two matrices, we must check their dimensions. For two matrices $A$ and $B$:

- Matrix $A$ has dimensions $(m \times n)$
- Matrix $B$ has dimensions $(p \times q)$

**Multiplication is only defined if** $n = p$ (the number of columns in $A$ must equal the number of rows in $B$). The resulting matrix $C$ will have dimensions $(m \times q)$.

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### 2. The General Formula

If $C = A \times B$, the element $c_{ij}$ is calculated as the sum of the products of the elements of the $i^{th}$ row of $A$ and the $j^{th}$ column of $B$:

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

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### 3. Step-by-Step Example

Let's multiply a $2 \times 3$ matrix by a $3 \times 2$ matrix.

**Given Matrices:**

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

**Calculation for Resulting Matrix $C$ $(2 \times 2)$:**

- **Element** $c_{11}$ (Row 1 · Col 1):

$$(1 \times 7) + (2 \times 9) + (3 \times 11) = 7 + 18 + 33 = 58$$

- **Element** $c_{12}$ (Row 1 · Col 2):

$$(1 \times 8) + (2 \times 10) + (3 \times 12) = 8 + 20 + 36 = 64$$

- **Element** $c_{21}$ (Row 2 · Col 1):

$$(4 \times 7) + (5 \times 9) + (6 \times 11) = 28 + 45 + 66 = 139$$

- **Element** $c_{22}$ (Row 2 · Col 2):

$$(4 \times 8) + (5 \times 10) + (6 \times 12) = 32 + 50 + 72 = 154$$

**Final Result:**

$$C = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}$$

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## 4. Properties to Remember

1. **Non-Commutative:** $AB \neq BA$ (usually).
2. **Associative:** $A(BC) = (AB)C$.
3. **Identity:** $AI = IA = A$.